

11. Credit Score Classification

Customer Loan Profile

```
from sklearn.tree import DecisionTreeClassifier
```

```
X = [[12,10,1,1],[10,8,2,1],[8,6,2,1],  
      [6,4,3,0],[4,2,4,0],[15,12,1,1],  
      [7,5,3,1],[5,3,4,0],[11,9,2,1],[3,1,5,0]]
```

```
y = ["Excellent","Good","Good","Average","Poor",  
      "Excellent","Average","Poor","Good","Poor"]
```

```
model = DecisionTreeClassifier(criterion="entropy", random_state=42)  
model.fit(X,y)
```

```
p = model.predict([[9,7,2,1]])  
print("Predicted Credit Score:", p[0])
```

OUTPUT

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit  
AMD64] on win32  
Enter "help" below or click "Help" above for more information.  
  
= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning  
lab/EXP 7.py  
Predicted Credit Score: Good  
|
```